

Module 4: A+ Troubleshooting and Helpdesk

DAY 5:-

Task 1:

Laptop Won't Power On (Even When Plugged In)

1. Verify the power source and outlet

Try a different wall outlet, and check the charging brick's LED indicator to confirm it's actually receiving power. Checking for whether the wall outlet or charger itself has failed.

2. Check for a charging LED on the laptop

Look for a small charging LED on the laptop itself when plugged in — if it lights up, the laptop is receiving power even if it won't boot. Checking for whether the issue is power delivery vs. the boot process itself.

3. Inspect the charging cable and port

Inspect the charger cable and DC jack (or USB-C port) for visible damage, bent pins, or looseness. Checking for a physical connection failure preventing power from reaching the laptop.

4. Perform a hard reset

Disconnect the charger, remove the battery if removable, hold the power button for 30 seconds, then reconnect only the charger (no battery) and try powering on. Checking for whether a faulty/drained battery is blocking the boot process.

5. Listen and watch for any signs of life

Listen for any fan spin, beep codes, or LED activity when pressing the power button. Checking for whether the system is actually attempting to boot (indicating a display/RAM issue) versus showing zero signs of life (pointing to a dead motherboard or power circuit).

Task 2:

Blank Screen, CPU Running

1. Check monitor power and cable connections

Confirm the monitor is actually powered on and the cable (HDMI/DisplayPort/VGA) is firmly connected at both the monitor and PC ends. Try a different cable or port if available.

2. Verify correct video output (GPU vs. motherboard)

If the PC has both integrated graphics and a dedicated GPU, confirm the monitor cable is plugged into the correct one — the GPU's ports if a dedicated card is installed, since motherboard video ports may be disabled or non-functional in that configuration.

3. Check the monitor's input source setting

If using multiple monitors or a docking station, check that the correct display input source is selected on the monitor itself (many monitors have physical buttons to cycle input sources).

4. Reseat the GPU if applicable

Power off, then confirm the GPU (if dedicated) is fully seated in its PCIe slot and its power connectors are attached. A loose GPU can allow the CPU/fans to run while producing no video output.

5. Listen for POST beeps or check diagnostic LEDs

Boot the PC and listen for POST beep codes or check for diagnostic LEDs on the motherboard. If POST fails at the video stage specifically, this points to a GPU or RAM fault rather than a cable/settings issue.

6. Test with integrated graphics (if available)

If a dedicated GPU is present and suspected faulty, remove it and connect the monitor directly to the motherboard's video output (if the CPU has integrated graphics) to test whether the system displays normally without it.

Task 3:

Wireless Mouse Unresponsive (Keyboard Works Fine):

Step 1: Rule out the mouse itself

- Check/replace the mouse's batteries.
- Turn the mouse off and on again (many wireless mice have a physical power switch)

Step 2: Rule out the USB receiver/port

- Try plugging the USB receiver into a different USB port on the same PC — if the mouse starts working, the original port was likely faulty or had a driver conflict.
- Test the receiver in a different computer if available — if it works there, the receiver itself is fine, pointing back to something specific about the original system.

Step 3: Rule out system/software settings

- Check **Device Manager** (Windows) for the mouse listed under "Mice and other pointing devices" — a yellow warning icon indicates a driver problem.
- Check if **Bluetooth or wireless connectivity is disabled** at the OS level (some laptops have a physical wireless toggle switch/key that could be accidentally switched off).
- Confirm the mouse isn't **paired to a different device** (common with Bluetooth mice that remember multiple paired devices).

Task 4:

Gaming Cafe: System Overheating and Shutting Down

Cause 1: Dust buildup blocking airflow

- **How to check:** Open the case and visually inspect the CPU heatsink, GPU fans, and case fans for dust accumulation — heavy dust buildup restricts airflow and traps heat
- **Fix:** Use compressed air to clean out dust from fans, heatsinks, and vents

Cause 2: Failing or insufficient case/CPU fans

- **How to check:** Listen for unusual fan noise (grinding, rattling) or watch whether fans are spinning at all when the system is under load; check fan speeds in BIOS or a monitoring tool
- **Fix:** Replace any fan that's not spinning properly or isn't keeping up with thermal load (common in high-traffic gaming cafe systems that run for many hours daily)

Cause 3: Dried-out or improperly applied thermal paste

- **How to check:** If dust cleaning and fan checks don't resolve it, this points to thermal paste — over time (especially with heavy daily use like a gaming cafe), thermal paste between the CPU/GPU and heatsink dries out and loses effectiveness, causing temperatures to climb even with working fans
- **Fix:** Remove the heatsink, clean off old thermal paste, and reapply a fresh, even layer before reseating the heatsink